Similarly, the following for loop also forms an empty loop.

```
int i;
for( i=0          ;          i<=5;    ++i);
{
        cout<<"\nWe are Indians";
}
```

The body of for loop contains a null statement

The body of for loop is not executed because semicolon(;) is ended at the end of for loop.

In the above code, the body of a for loop enclosed by braces is not executed at all because a semicolon is ended after the for loop.

Declaration of variable in a for loop

In C++, the variables can also be declared within a for loop. For instance,

```
int main ()
{
   int sum = 0;
   for(int i=0;  i<=5;  ++i )
   {
        sum = sum + i;
   }
   cout<<"\nThe variable i cannot access here";
   cout<<"\n The variable sum can access here";
}
```

Variable (i)is declared within the for loop.

The variable i can be accessed only within the body of loop.

A variable declared inside the block of main() can be accessed anywhere inside main() i.e., the scope of variable in main()

10.5.3 While loop

A while loop is a control flow statement that allows the loop statements to be executed as long as the condition is true. The while loop is an entry-controlled loop because the test-expression is evaluated before the entering into a loop.

The while loop syntax is:

```
while ( Test expression )
{
        Body of the loop;
}
Statement-x;
```